

# Daniel Hardesty Lewis

*Causal AI & Staff ML Engineer – causal inference, double ML, spatiotemporal forecasting*

Specializing in causal inference, spatiotemporal forecasting, and robust validation frameworks. Developed novel invariant causal learning algorithms, real-time geographic forecasting systems, and statewide disaster resilience platforms.

## Core Expertise

- 8 years **Causal AI & ML Engineering:** Build and operate structural causal models, counterfactual inference pipelines, and ML systems using rigorous validation protocols to ensure OOD generalization and reliability
- 8 years **Backend & Data:** Build schemas and ETL; ship API-backed services using Python, SQL, and Postgres/PostGIS
- 8 years **Distributed Systems:** Scale batch workloads across distributed systems, profile bottlenecks, and optimize latency & throughput with explicit cost-performance tradeoffs
- 10 years **Geospatial Causal AI:** Spatiotemporal causal inference, double machine learning, and graphical modeling for risk, planning, and hazard decision-support
- 8 years **Technical Leadership:** Drive cross-functional execution via specifications and governance; mentor engineers through design reviews, code reviews, and instruction

## Technologies

|                |                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------|
| Causal AI      | DoWhy; EconML; PyWhy; CausalNex; structural causal models; counterfactual inference; do-calculus |
| Modeling       | PyTorch; TensorFlow; scikit-learn; Transformers; Double ML; Causal Forests; tree ensembles       |
| Data stack     | Python; SQL; Pandas; Polars; PyArrow; Parquet; Postgres; PostGIS; raster/vector GIS toolchains   |
| Infrastructure | Linux; Bash; Git; Docker; Kubernetes; Ray; Dask; CUDA; Modulus; Earth-2; Triton Inference Server |

## Experience Professional

- Jun. 2026 – Present **Member Resident**, [South Park Commons](#), New York, NY  
Exploring spatiotemporal foundation models and causal inference for infrastructure decision-support, focused on AI data center feasibility and risk quantification.
- Apr. 2025 – Present **Founder**, [Homecastr](#), New York, NY  
Consumer AI forecasting platform serving thousands of monthly active users.
  - Architected a nationwide neighborhood-level diffusion forecasting model, matching Zillow's 7.3% 0-year error benchmark at a 1-year horizon and maintaining 35.6% median error over 4 years.
  - Engineered architecture with spatial inducing tokens, per-horizon loss normalization, and DDIM sampling.
  - Developed predictive validation frameworks optimizing PIT uniformity and interval sharpness; owned end-to-end pipeline from data ingestion to real-time serving.
- Oct. 2024 – **Technical Advisor**, [PoliBOM](#), Seattle, WA
- Sep. 2025 AI tariff mitigation and trade compliance product for manufacturers.
  - Engineered an agentic RAG pipeline for BOM parsing and tariff simulation, replacing manual workflows with multi-step reasoning.
  - Directed development of conversational interfaces for proactive compliance alerts; presented prototype to KPMG.
- Nov. 2023 – **Founder**, [Summit Geospatial](#), New York, NY
- Mar. 2025 High-resolution elevation data and web delivery for engineering and hazard workflows.
  - Engineered a **statewide seamless Texas DEM mosaic**, resampling 0.5 m LiDAR sources to 1.2 m via nearest-neighbor across **70+ state, federal, and local elevation datasets**.
  - Developed the web distribution platform end-to-end including **data pipeline, tiling, hosting, delivery UX** to support engineering and planning use cases.
- Aug. 2021 – **Senior Data Scientist & Technical Lead**, [TACC](#), Austin, TX
- Aug. 2023
  - Commercialized state-level terrain models from the \$40M [Texas Disaster Information System \(TDIS\)](#), spinning out Summit Geospatial alongside UT's VPs of IP.
  - Scaled spatiotemporal climate and flood models on leadership-class supercomputers, executing 800k-node distributed jobs and managing \$1M+ compute allocations (NOAA, NSF, USACE).
  - Partnered with USACE to statistically model compound coastal flood hazards, integrating high-resolution topographic data with hydrological models.
  - Served as Technical Reviewer for Frontera Doctoral Fellows and Large-Scale Computing Pathways applications, evaluating resource requests on world-class supercomputing infrastructure.
- Feb. 2018 – **Data Scientist & Research Engineer**, [TACC](#), Austin, TX
- Jul. 2021
  - Competed in [DARPA World Modelers](#) to improve disaster resiliency and food security modeling.
  - Mentored industry partners in deep learning/HPC; received research recognition from AAAI President Yolanda Gil for automated scientific modeling.

## Research

- Sep. 2024 – **Graduate Thesis Researcher**, [Columbia University GSAPP](#), New York, NY
- May 2026
- Formulated Spatially-Weighted Invariant Risk Minimization (S-IRM) in PyTorch to extract robust spatial features.
  - Developed Invariant Double Machine Learning (I-DML) chaining ICP pre-filtering to eliminate spatial wealth bias.
  - Designed Adversarial Causal Spatial Representation Learning (Adversarial Autoencoder) to isolate spatial geometries.
  - Evaluated architectures on 193k case-period panel, backtesting causal estimates (E-value: 1.49).
- May 2025 – **Research assistant to Dir. Electrical Engineering Zoran Kostic**, [Columbia University](#), New York, NY
- Aug. 2025
- Math benchmarking and finetuning of Deepseek’s GRPO algorithm
- Jan. 2025 – **Research assistant to Dir. Financial Engineering Ali Hirsra (Industry research)**, [Columbia University](#), New York, NY
- Oct. 2025
- Developed multi-asset CVAE factor model with Skew-T Mixture priors (85%  $R^2$ ); derived a tractable SHAP algorithm using Hessian-based recursive clustering for deep learning attributions.
- July 2019 – **Summer research intern to Pres. AAI Yolanda Gil**, [University of Southern California](#), Los Angeles, CA
- Aug. 2019
- Integrated hydrological models into the [MINT Platform](#) during a year-long competition with MITRE to provide DARPA the "Airflow for science"

## Teaching

- Jan. 2018 – **Co-instructor**, [The University of Texas at Austin](#), Austin, TX
- Dec. 2020
- Helped design and teach graduate-level courses: *Machine Learning for the Geosciences* for Petrobras engineers, *Intelligent Systems for the Geosciences*, and *Scientific Computation*.

## Select Publications

- 2021 Gil, Y., et al. (incl. **D. Hardesty Lewis**), “Artificial Intelligence for Modeling Complex Systems: Taming the Complexity of Expert Models to Improve Decision Making”. *ACM TüS*, 2021. DOI: [10.1145/3453172](#)
- 2019 Garijo, D., et al. (incl. **D. Hardesty Lewis**), “An intelligent interface for integrating climate, hydrology, agriculture, and socioeconomic models”. *ACM IUI '19*, 2019. DOI: [10.1145/3308557.3308711](#)

## Education

- expected 2026 **M.S., Urban Planning**, [Columbia University](#), New York City, NY
- Concentration in Spatial Analytics. Thesis: *Predicting Formal Protest Petitions: Evidence from Austin, Texas* (Causal Inference & DML).
- 2017 **B.S., Pure Mathematics**, [University of Texas](#), Austin, TX

## Key Projects

- oppsAlert Causal AI and predictive forecasting pipeline; integrates a 193k case-period spatial panel with TCAD appraisal rolls and ACS demographics using Double Machine Learning (DML) and tree ensembles (CatBoost, XGBoost).
- M&A predictor Built M&A prediction model (Oct.–Dec. 2025) with Prof. Eric Talley on 1.5M Compustat firm-quarters; long-only portfolio beat S&P 500 by 2%/yr (2004–2020), peaking at 6%/yr.

## Activities

- Sailing Volunteer Atlantic SAR of the [Cheeki Rafiki](#)      Bikepacking US West & East Coasts

## Languages

- Spanish Native Proficiency      French Professional Working Proficiency